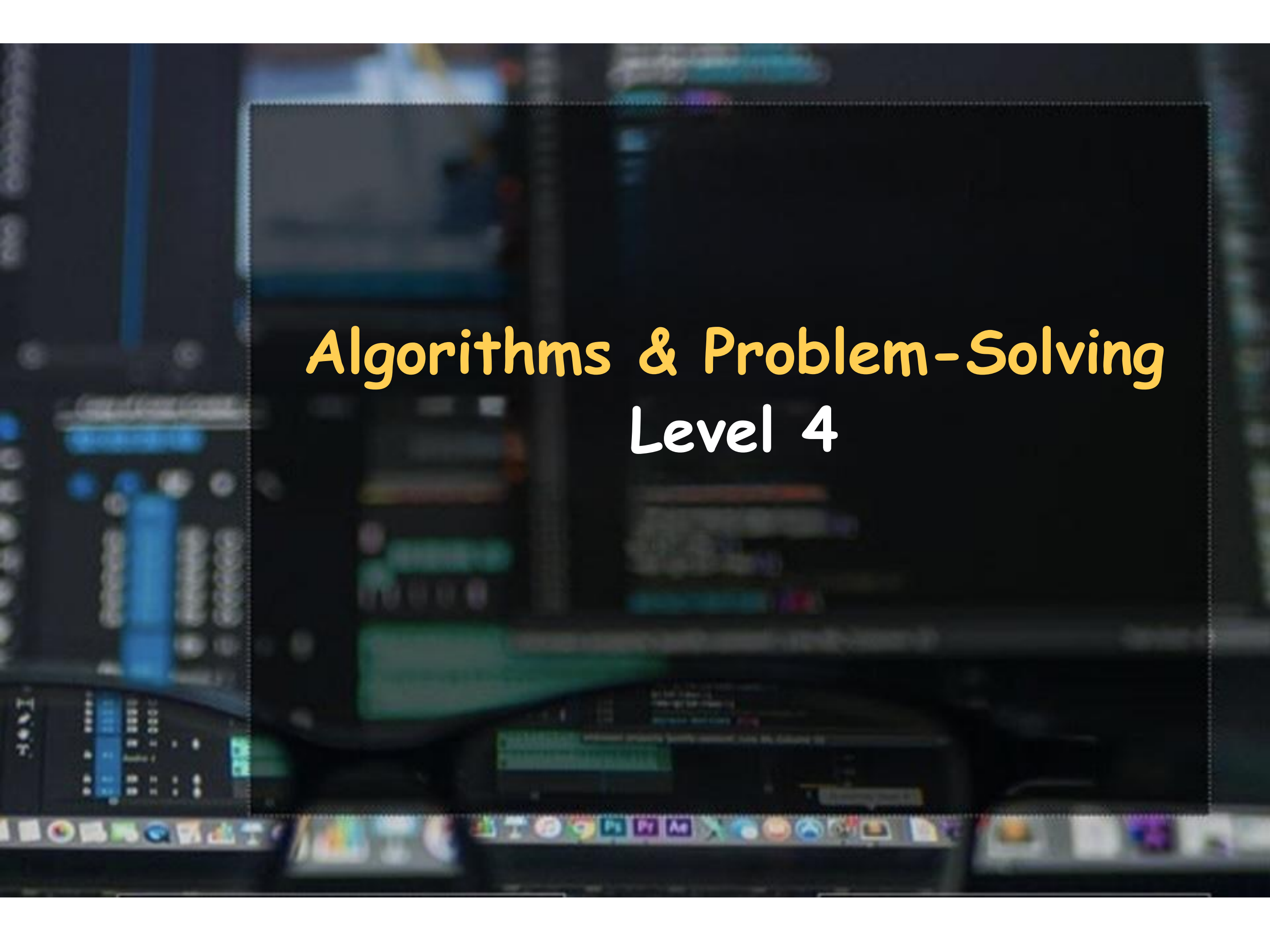
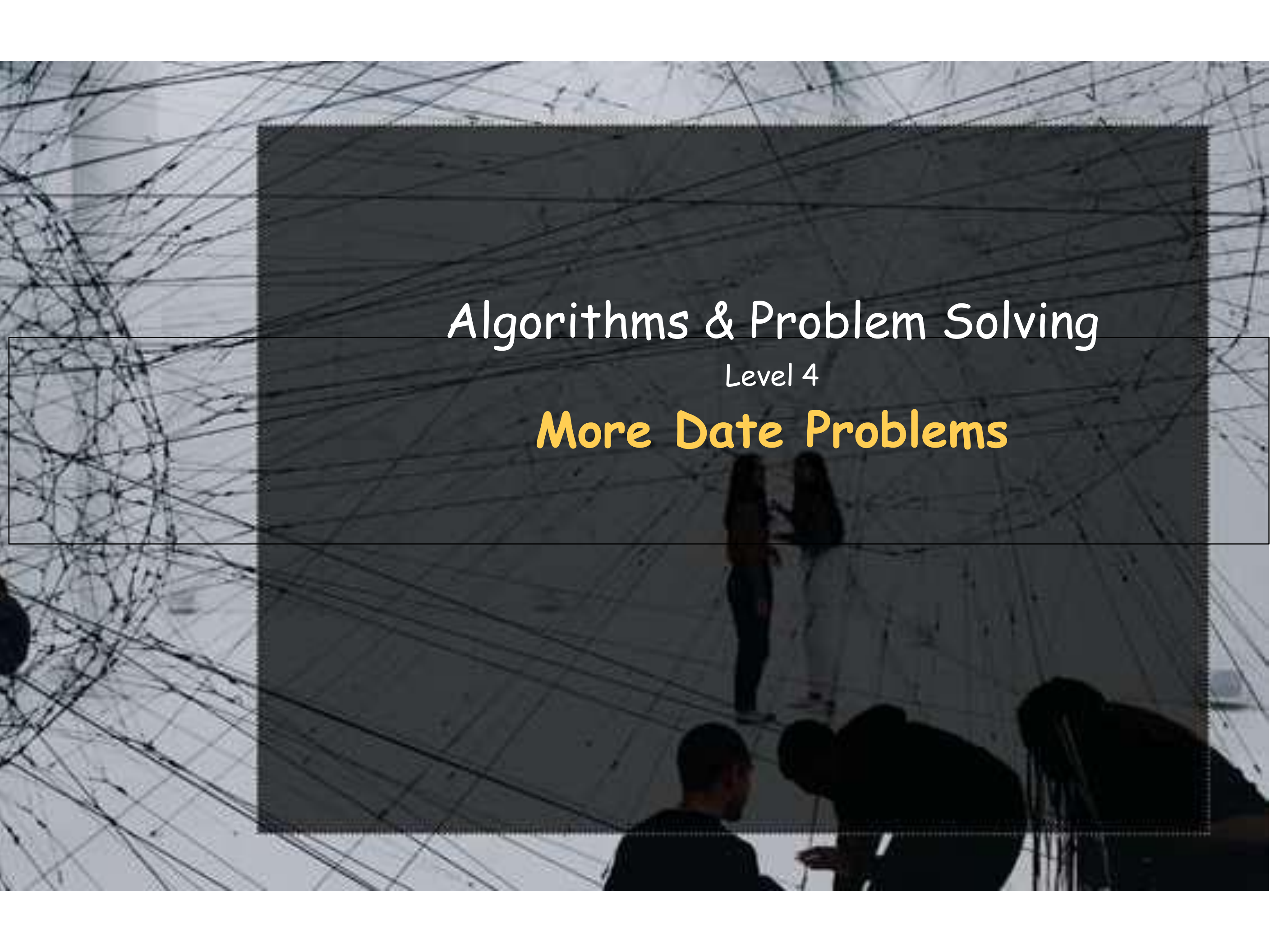




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

More Date Problems



Definition

- The goal is to create a C++ program that reads a date (day, month, year) and calculates a set of date-related information such as:
 - Day of the week
 - Is it a weekend or a workday?
 - And the number of days remaining until the end of the week, month, and year.



Explanation:

■ The goal is a program that uses a structure to store dates:

```
struct sDate
```

```
{
```

```
    short Year;
```

```
    short Month;
```

```
    short Day;
```

```
};
```

Procedures

- 1 Reading the date from the user
 -  Contains ReadDay(), ReadMonth(), ReadYear(), and ReadFullDate().
These functions read the day, month, and year and return an sDate object.
- 2 Checking for a leap year
 -  IsLeapYear(year)
 -  Returns true if the year is a leap year (divisible by 4, not 100 unless it's divisible by 400).
- 3 Number of days in a month
 -  NumberOfDaysInAMonth(month, year)
 -  Used to check the end of the month.
- 4 Calculating the day of the week
 -  DayOfWeekOrder(day, month, year)
 -  Returns the day of the week number (0=Sunday, 6=Saturday).
 -  DayShortName() Returns the abbreviated name (e.g., Sun, Mon, ...).
- 5 Checking Day Type
 -  IsEndOfWeek(date) Is it Saturday?
 -  IsWeekEnd(date) Is today Friday or Saturday?
 -  IsBusinessDay(date) → Is today Sunday to Thursday?
- 6 Increase the date by one day
 -  IncreaseDateByOneDay(date)
 -  Used to calculate the difference between two dates

7 Calculating Time Differences

- `GetDifferenceInDays(date1, date2)`

- Returns the number of days between two dates

8 Days Until the End of the Week/Month/Year

- `DaysUntilTheEndOfWeek(date)`

- `DaysUntilTheEndOfMonth(date)`

- `DaysUntilTheEndOfYear(date)`



Example:

Today is Fri, 23/9/2022

Is it End of Week?

No Not end of week.

Is it Weekend?

Yes it is a week end.

Is it Business Day?

No it is NOT a business day.

Days until end of week : 1 Day(s).

Days until end of month : 8 Day(s).

Days until end of year : 100 Day(s).



Note:

- The program can automatically use the current system date or manually read the date from the user.

Summary

- ✓ The program applies practical concepts in date management such as:
 - Date validity verification
 - Date calculations
 - Day type determination (work/holiday)
 - Using multiple functions for logical segmentation

Thank you 😊